

Derivation of the Euler–Maruyama Scheme

Non-affine Stochastic-Volatility Model

Based on the MTH321 Problem Pack, Direction 4

1 The model to be approximated

The non-affine model is written in the log-price variable

$$X_t = \log S_t,$$

with volatility factor Y_t :

$$dX_t = \left(\mu - \frac{1}{2}g(Y_t)^2 \right) dt + g(Y_t) dW_t^{(1)}, \quad (1)$$

$$dY_t = \kappa(\theta - Y_t)dt + \xi \sqrt{1 + Y_t^2} dW_t^{(2)}, \quad (2)$$

where

$$g(y) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}},$$

and the two Brownian motions satisfy

$$d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt, \quad |\rho| < 1.$$

After computing X_t , the asset price is recovered by

$$S_t = e^{X_t}.$$

The numerical task is therefore to approximate the coupled pair (X_t, Y_t) . The Euler–Maruyama method is applied directly to these two SDEs.

2 General Euler–Maruyama derivation

Start from a scalar SDE

$$dZ_t = a(Z_t) dt + b(Z_t) dW_t.$$

Integrate over one time interval $[t_n, t_{n+1}]$, where

$$h = t_{n+1} - t_n.$$

Then

$$Z_{t_{n+1}} - Z_{t_n} = \int_{t_n}^{t_{n+1}} a(Z_s) ds + \int_{t_n}^{t_{n+1}} b(Z_s) dW_s.$$

Euler–Maruyama freezes the coefficients at the left endpoint:

$$a(Z_s) \approx a(Z_n), \quad b(Z_s) \approx b(Z_n).$$

Hence

$$\int_{t_n}^{t_{n+1}} a(Z_s) ds \approx a(Z_n)h,$$

and

$$\int_{t_n}^{t_{n+1}} b(Z_s) dW_s \approx b(Z_n)(W_{t_{n+1}} - W_{t_n}).$$

Define the Brownian increment

$$\Delta W_n = W_{t_{n+1}} - W_{t_n}.$$

Therefore the Euler–Maruyama update is

$$Z_{n+1} = Z_n + a(Z_n)h + b(Z_n)\Delta W_n.$$

Because Brownian increments satisfy

$$\Delta W_n \sim N(0, h),$$

we may generate $Z_n^* \sim N(0, 1)$ and write

$$\Delta W_n = \sqrt{h} Z_n^*.$$

3 Euler–Maruyama update for X_t

For the first equation,

$$dX_t = \left(\mu - \frac{1}{2}g(Y_t)^2 \right) dt + g(Y_t)dW_t^{(1)},$$

the drift and diffusion coefficients are

$$a_X(X, Y) = \mu - \frac{1}{2}g(Y)^2, \quad b_X(X, Y) = g(Y).$$

Integrating over $[t_n, t_{n+1}]$ gives

$$\begin{aligned} X_{t_{n+1}} - X_{t_n} &= \int_{t_n}^{t_{n+1}} \left(\mu - \frac{1}{2}g(Y_s)^2 \right) ds \\ &\quad + \int_{t_n}^{t_{n+1}} g(Y_s) dW_s^{(1)}. \end{aligned}$$

Freeze the coefficient at Y_n :

$$g(Y_s) \approx g(Y_n).$$

Then

$$\int_{t_n}^{t_{n+1}} \left(\mu - \frac{1}{2}g(Y_s)^2 \right) ds \approx \left(\mu - \frac{1}{2}g(Y_n)^2 \right) h,$$

and

$$\int_{t_n}^{t_{n+1}} g(Y_s) dW_s^{(1)} \approx g(Y_n)\Delta W_n^{(1)}.$$

Therefore

$$X_{n+1} = X_n + \left(\mu - \frac{1}{2}g(Y_n)^2 \right) h + g(Y_n)\Delta W_n^{(1)}.$$

4 Euler–Maruyama update for Y_t

For the second equation,

$$dY_t = \kappa(\theta - Y_t)dt + \xi\sqrt{1 + Y_t^2} dW_t^{(2)},$$

the coefficients are

$$a_Y(Y) = \kappa(\theta - Y), \quad b_Y(Y) = \xi\sqrt{1 + Y^2}.$$

Over one step,

$$\begin{aligned} Y_{t_{n+1}} - Y_{t_n} &= \int_{t_n}^{t_{n+1}} \kappa(\theta - Y_s) ds \\ &\quad + \int_{t_n}^{t_{n+1}} \xi\sqrt{1 + Y_s^2} dW_s^{(2)}. \end{aligned}$$

Freeze the coefficient at Y_n :

$$\begin{aligned} \kappa(\theta - Y_s) &\approx \kappa(\theta - Y_n), \\ \xi\sqrt{1 + Y_s^2} &\approx \xi\sqrt{1 + Y_n^2}. \end{aligned}$$

Therefore

$$Y_{n+1} = Y_n + \kappa(\theta - Y_n)h + \xi\sqrt{1 + Y_n^2} \Delta W_n^{(2)}.$$

5 Constructing the correlated Brownian increments

The two Brownian motions are not independent. They must have correlation ρ . Generate two independent standard normal variables at every time step:

$$Z_{1,n}, Z_{2,n} \stackrel{\text{iid}}{\sim} N(0, 1).$$

Define

$$\Delta W_n^{(1)} = \sqrt{h} Z_{1,n},$$

$$\Delta W_n^{(2)} = \sqrt{h} \left(\rho Z_{1,n} + \sqrt{1 - \rho^2} Z_{2,n} \right).$$

Why this construction is correct

For the first increment,

$$E[\Delta W_n^{(1)}] = 0, \quad \text{Var}(\Delta W_n^{(1)}) = h.$$

For the second,

$$\begin{aligned} \text{Var}(\Delta W_n^{(2)}) &= h \text{Var} \left(\rho Z_{1,n} + \sqrt{1 - \rho^2} Z_{2,n} \right) \\ &= h (\rho^2 + 1 - \rho^2) = h. \end{aligned}$$

Their covariance is

$$\begin{aligned} \text{Cov}(\Delta W_n^{(1)}, \Delta W_n^{(2)}) &= h E \left[Z_{1,n} \left(\rho Z_{1,n} + \sqrt{1 - \rho^2} Z_{2,n} \right) \right] \\ &= h \rho, \end{aligned}$$

because $E[Z_{1,n}^2] = 1$ and $E[Z_{1,n}Z_{2,n}] = 0$. Thus

$$\text{Corr}(\Delta W_n^{(1)}, \Delta W_n^{(2)}) = \frac{\rho h}{\sqrt{h}\sqrt{h}} = \rho.$$

So both increments have the correct Brownian variance and the required correlation.

6 Final Euler–Maruyama scheme

Substitute the normal-variable representation directly into the two updates. At each time step n ,

$$g_n = g(Y_n) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-Y_n}}.$$

Then

$$\begin{aligned} X_{n+1} &= X_n + \left(\mu - \frac{1}{2}g_n^2 \right) h + g_n \sqrt{h} Z_{1,n}, \\ Y_{n+1} &= Y_n + \kappa(\theta - Y_n)h \\ &\quad + \xi \sqrt{1 + Y_n^2} \sqrt{h} \left(\rho Z_{1,n} + \sqrt{1 - \rho^2} Z_{2,n} \right), \\ S_{n+1} &= e^{X_{n+1}}. \end{aligned}$$

This is the numerical Euler–Maruyama approximation used for the non-affine model.

7 Initial values for the reproducible benchmark

The Problem Pack specifies

$$S_0 = 100, \quad Y_0 = 0.$$

Since $X_0 = \log S_0$,

$$X_0 = \log 100.$$

The remaining parameters are

$$\begin{aligned} \mu &= 0.05, \quad \kappa = 2, \quad \theta = -0.2, \quad \xi = 0.6, \\ \rho &= -0.7, \quad \sigma_{\min} = 0.10, \quad \sigma_{\max} = 0.50, \quad T = 1. \end{aligned}$$

For a chosen step size $h = T/N$, repeat the update for

$$n = 0, 1, \dots, N - 1.$$

One complete sequence of random pairs $(Z_{1,n}, Z_{2,n})$ produces one Monte Carlo path.

One-step computational order:

1. Evaluate $g(Y_n)$.
2. Generate independent $Z_{1,n}, Z_{2,n} \sim N(0, 1)$.
3. Construct correlated $\Delta W_n^{(1)}$ and $\Delta W_n^{(2)}$.
4. Update X_{n+1} and Y_{n+1} with Euler–Maruyama.
5. Recover $S_{n+1} = e^{X_{n+1}}$.